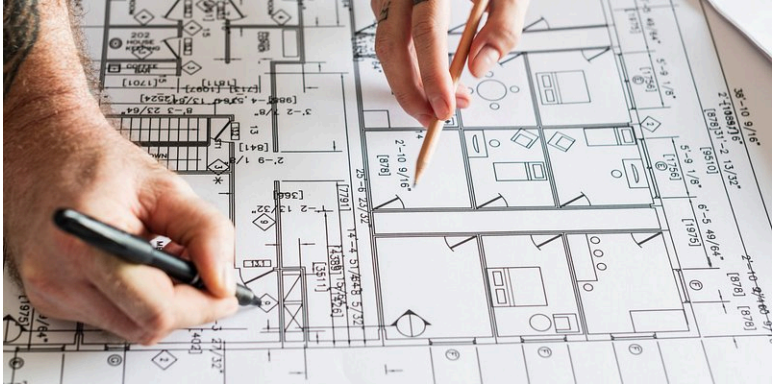




INTRODUCTION TO DATA MODELLING IN THE HUMANITIES



Information management

Information architecture

Information systems

MODELS

- A **computational model** is a representation of the object of study that is understandable by a computer.

MODELS

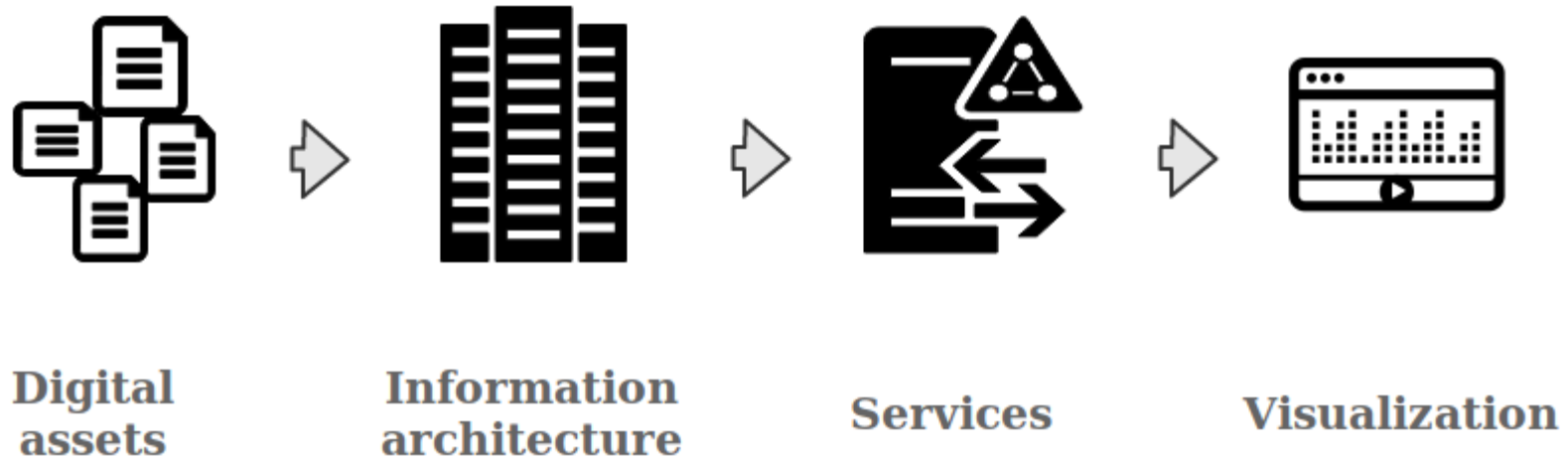
- A **computational model** is a representation of the object of study that is understandable by a computer.
- A **data model**, in particular, is a structured representation of the object of study that organise and describe the knowledge about that object.

MODELS

- A **computational model** is a representation of the object of study that is understandable by a computer.
- A **data model**, in particular, is a structured representation of the object of study that organise and describe the knowledge about that object.
- **Statistical models**, used in machine learning, learn from data and are not human-interpretable.

Primary sources in the humanities: manuscripts, diaries, letters, speeches, memoirs, historical documents, government records, works of art, music, films, photographs, architecture, performances, artifacts, objects, archival materials, newspaper articles, etc.

INFORMATION ARCHITECTURE



Intro to Digital Humanities. [1B: Analysis of DH projects](#). Johanna Drucker, UCLA



**Digital
assets**



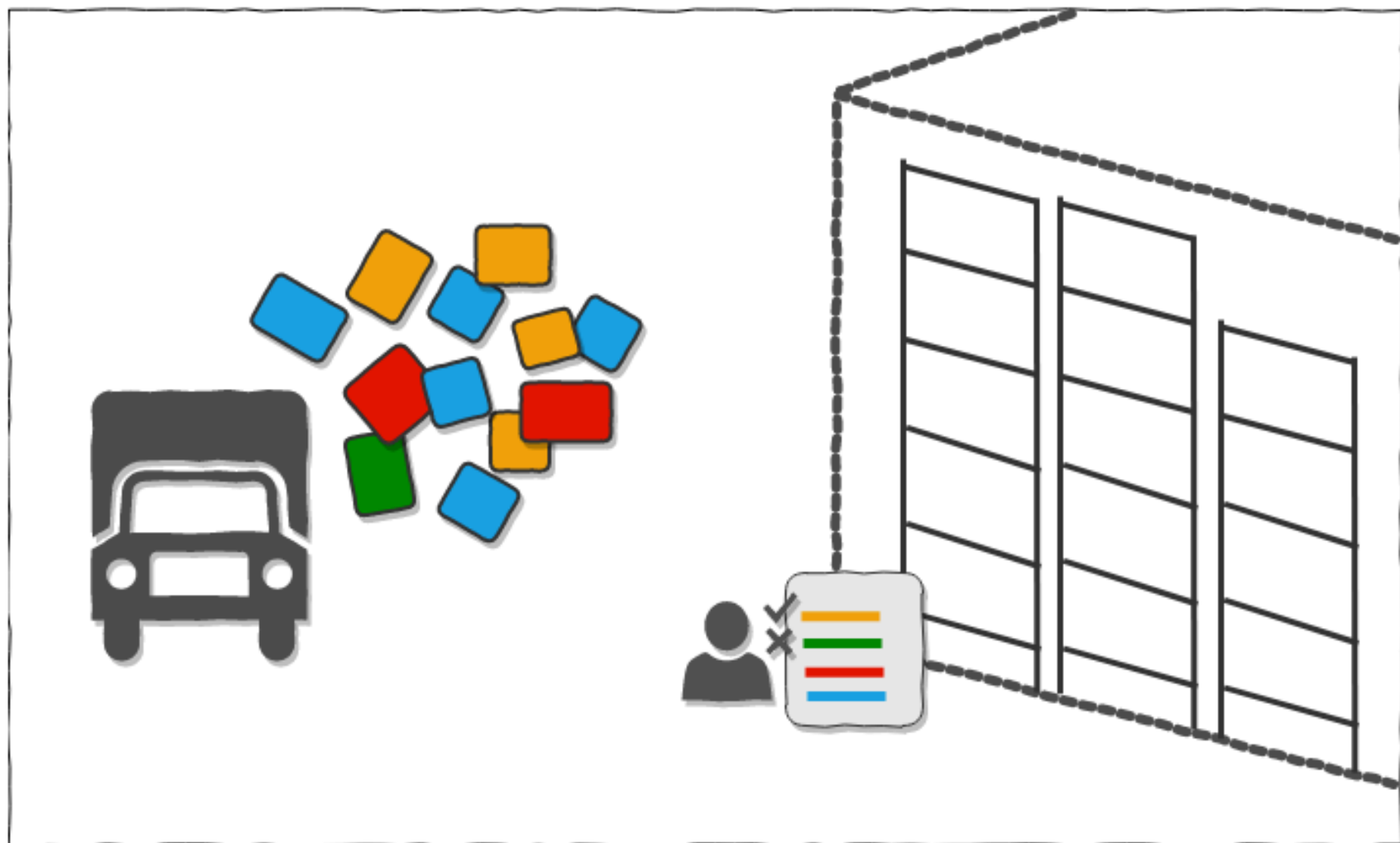
**Information
architecture**

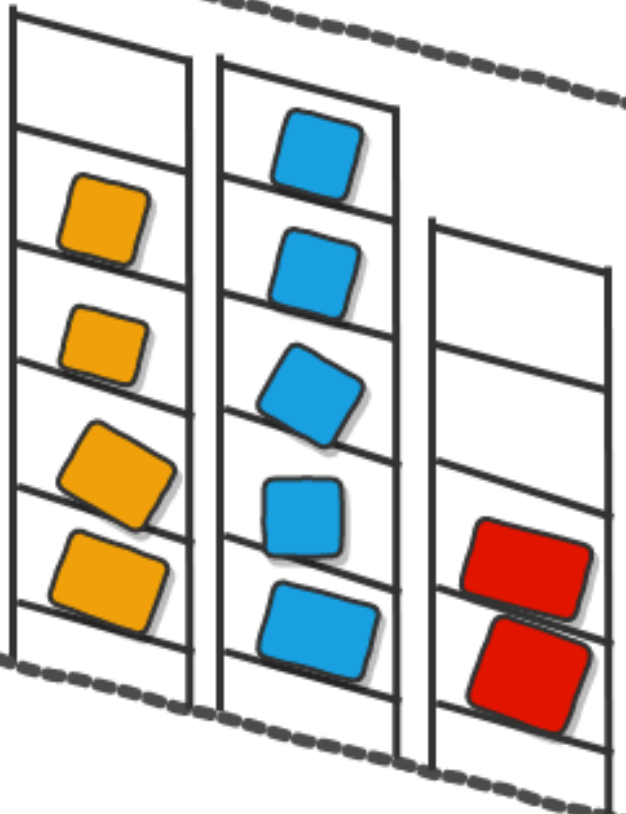


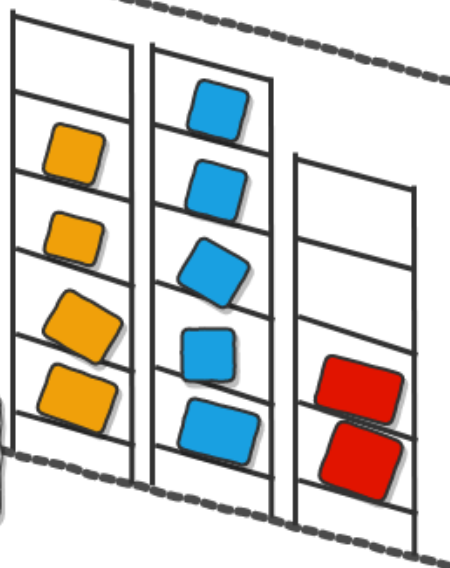
Services



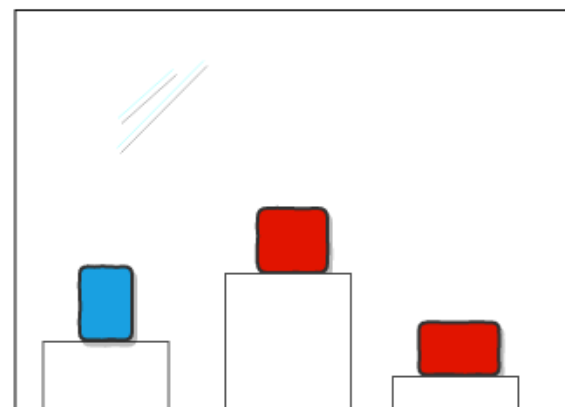
Visualization

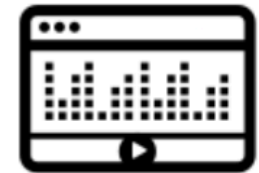
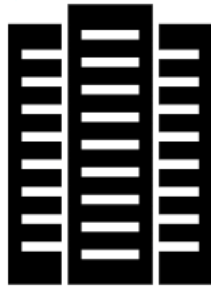






Scatole





Digital
assets

Information
architecture

Services

Visualization

warehouse / back of the shop

salesperson

shop window

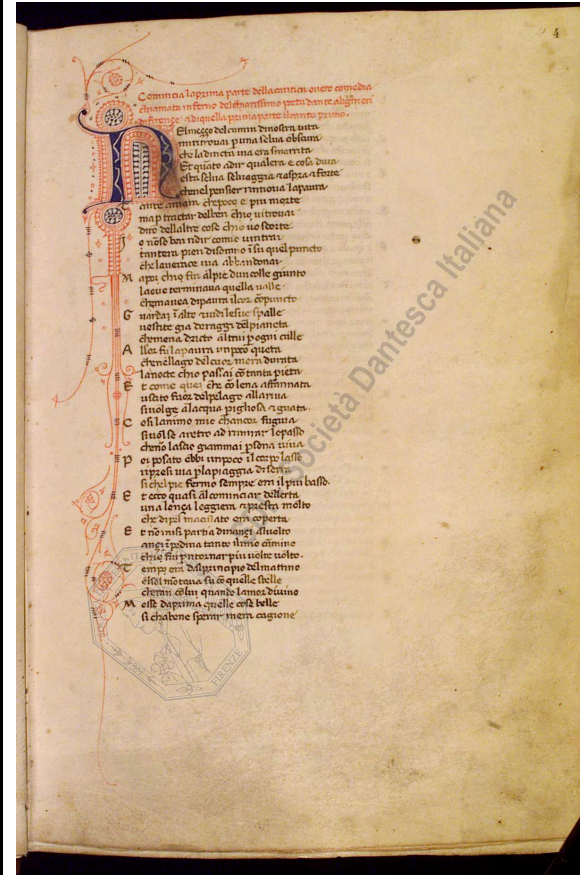
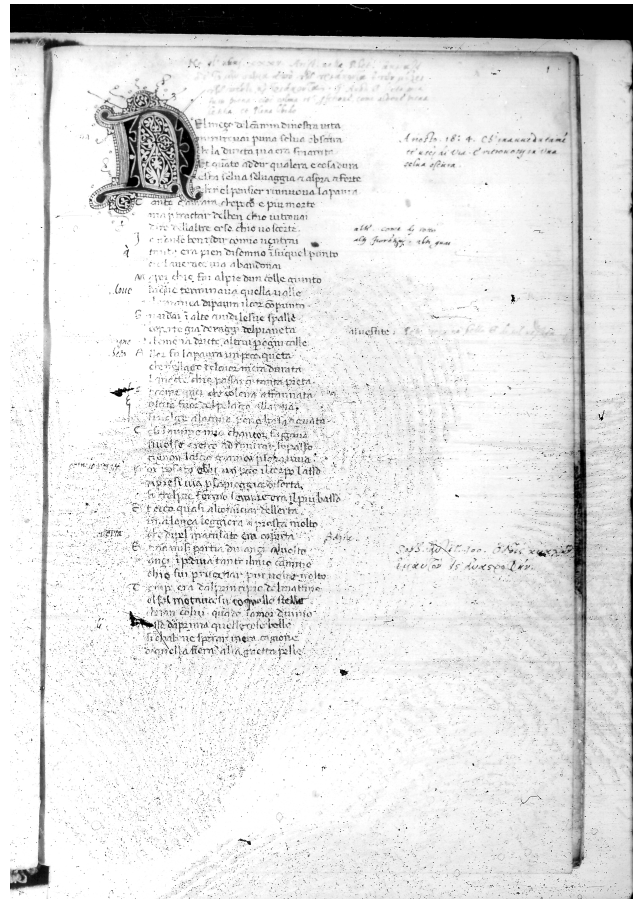
EXAMPLE



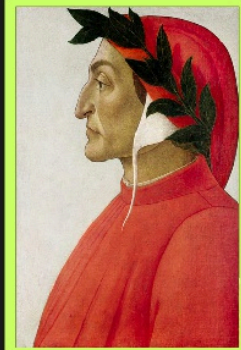
La Commedia di Boccaccio

See <https://journal.digitalmedievalist.org/article/id/7019/>

Riccardiano | Chigiano | Toledano

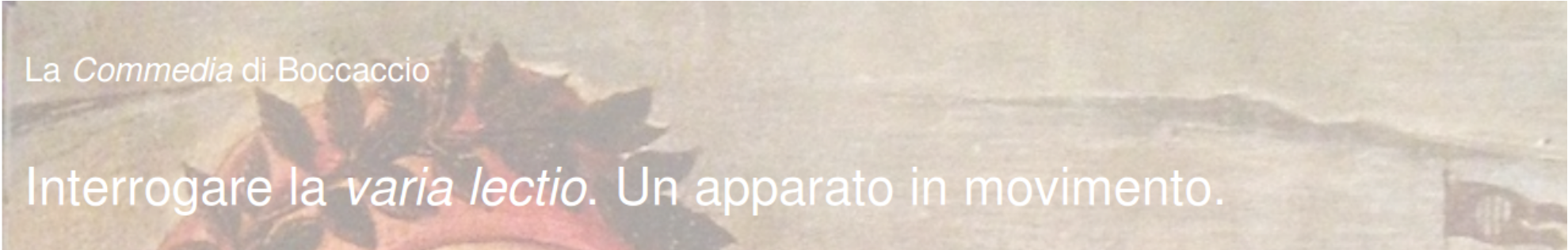


Brainstorming: how to describe them? Which of their characteristics, qualities, features, aspects, information are relevant for describing and representing them, for example in a db?



Antica Vulgata





La *Commedia* di Boccaccio

Interrogare la *varia lectio*. Un apparato in movimento.

[Per visualizzare tutte le varianti, clicca qui](#)

Cerca

<https://boccacciocommedia.it>

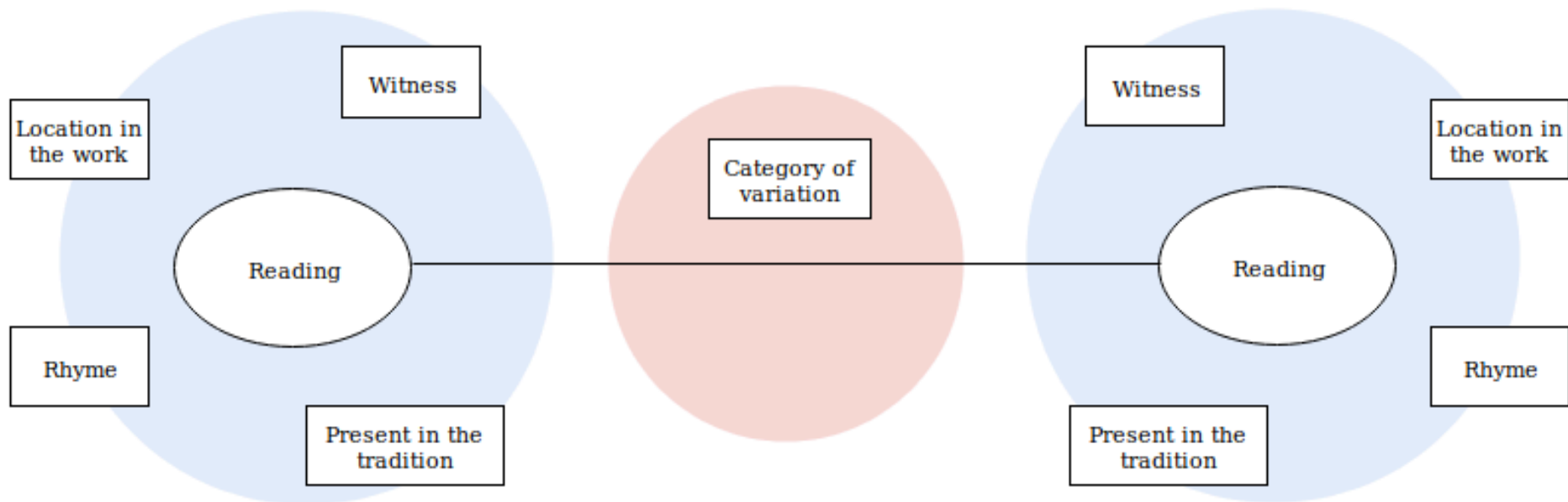
[Per raffinare la ricerca, usare uno o più criteri](#)

Da cantica e canto:

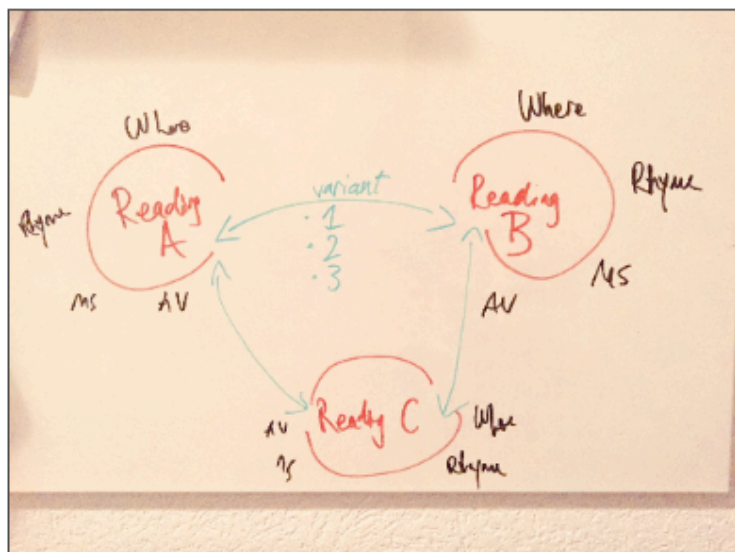
A cantica e canto:

Scegli testimoni da oppore:

In rima:



conceptual



logical



OWL



**ODD
(XML/TEI)**



SQL

1



modello concettuale

2

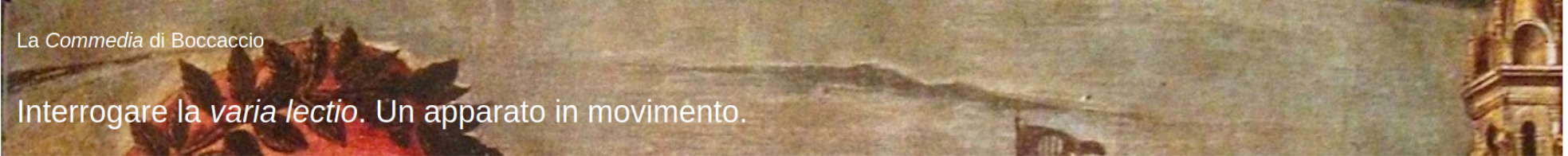


modello logico

3

The interface features a header with the title 'La Commedia di Boccaccio' and the subtitle 'Interrogare la *vara lectio*. Un apparato in movimento.' Below the header, there is a search bar with a 'Cerca' button. A message states: 'Per visualizzare tutte le varianti, clicca qui'. Another message says: 'Per raffinare la ricerca, usare uno o più criteri'. There are two input fields for 'Da cambiare e sento:' and 'A cambiare e sento:'. Below these, there are two input fields for 'Scegli testimoni da opporre:' and 'In rima:'.

implementazione



La *Commedia* di Boccaccio

Interrogare la *varia lectio*. Un apparato in movimento.

Per visualizzare tutte le varianti, clicca qui

Cerca

Per raffinare la ricerca, usare uno o più criteri

Da cantica e canto:

A cantica e canto:

Scegli testimoni da opporre:

In rima:

Assente nella tradizione precedente i codici del Boccaccio:

Tipo di variante (quando assente nella tradizione):

Cerca

Risultati: 17

In **azzurro** le lezioni assenti nella tradizione precedente a Boccaccio.

Mostra

25

 risultati per pagina

Cerca:

			Toledano	Riccardiano	Chigiano	Tipo di variante	In rima	Ed. Petrocchi	
Inf	I	50	magreçça	magreçça	grameça	eco / anticipazione	in rima	sembiava carca ne la sua magrezza	Nota
Inf	I	52	graveçça	graveçça	grameça	eco / anticipazione	in rima	questa mi porse tanto di gravezza	Nota
Inf	I	89	saggio	saggio	et saggio		in rima	aiutami da lei, famoso saggio ,	
Inf	II	7	maitate	maiutate	maiutate		in rima	O muse, o alto ingegno, or m'aiutate ;	
Inf	II	90	poderose	paurose	paurose	lessicale	in rima	de l'altre no, ché non son paurose .	Nota
Inf	IV	29	molte et grandi	molte et grandi	molto grandi		in rima	ch'avean le turbe, ch'eran molte e grandi ,	
Inf	V	57	conducta	condotta	corrotta		in rima	per tòrre il biasmo in che era condotta .	Nota
Inf	VII	119	summo	sommo	sommo		in rima	e fanno pullular quest'acqua al summo	
Inf	IX	45	trine	crine	crine		in rima	«Guarda», mi disse, «le feroci Erine .	Nota
Inf	IX	70	fiori (ad marg. fuori)	fuori	fuori		in rima	li rami schianta, abbatte e porta fori ;	
Inf	IX	117	amaro	amaro	avaro	eco / anticipazione	in rima	salvo che 'l modo v'era più amaro	Nota
Inf	XXXI	116	hereda	reda	reda		in rima	che fece Scipion di gloria reda ,	
Inf	XXXII	16	scuro	oscuro	oscuro		in rima	Come noi fummo giù nel pozzo scuro	
Inf	XXXIII	59	per voglia	per voglia	per fame	lessicale	in rima	ed ei, pensando ch'io 'l fessi per voglia	Nota
Inf	XXXIV	114	consunto	consuncto	congiunto	eco / anticipazione	in rima	coverchia, e sotto 'l cui colmo consunto	Nota
Par	I	78	iscerni	discerni	discerni		in rima	con l'armonia che temperi e discerni ,	
Par	IV	112	expreme	spreme	spreme		in rima	Però, quando Piccarda quello spreme ,	

Per raffinare la ricerca, usare uno o più criteri

Da cantica e canto:

A cantica e canto:

Scegli testimoni da oppore:

In rima:

Presente nella tradizione precedente i codici del Boccaccio:

Tipo di variante (quando assente nella tradizione):

Cerca

Risultati: 6

In [azzurro](#) le lezioni assenti nella tradizione precedente a Boccaccio.

Mostra

25

 risultati per pagina

Cerca:

			◀◀◀ Toledano ▶▶▶	◀◀◀ Riccardiano ▶▶▶	◀◀◀ Chigiano ▶▶▶	Tipo di variante	◀◀◀ In rima ▶▶▶	Ed. Petrocchi	◀◀◀ ▶▶▶
Inf	IV	141	tullio et lyno	tulio lino	tulio lino	inversione	non in rima	Tulïo e Lino e Seneca morale;	
Inf	VIII	24	tal si fece	fecesi tal	fecesi tal	inversione	non in rima	fecesi Flegïas ne l'ira accolta	Nota
Inf	VIII	111	che si et no (ad marg. al. chel si el no)	chel no el si	chel no el si	inversione	non in rima	che si e no nel capo mi tenciona	
Inf	IX	75	ove e quel fummo piu	onde quel fummo e piu	onde quel fummo e piu	inversione	non in rima	per indi ove quel fummo è più acerbo	
Inf	XXXII	135	dissio perche	perche dissio	perche dissio	inversione	non in rima	dimmi 'l perché", diss'io, "per tal convegno,	
Par	IV	47	gabriel et michel	michele et gabriel	michele et gabriel	inversione	non in rima	Gabriel e Michel vi rappresenta,	

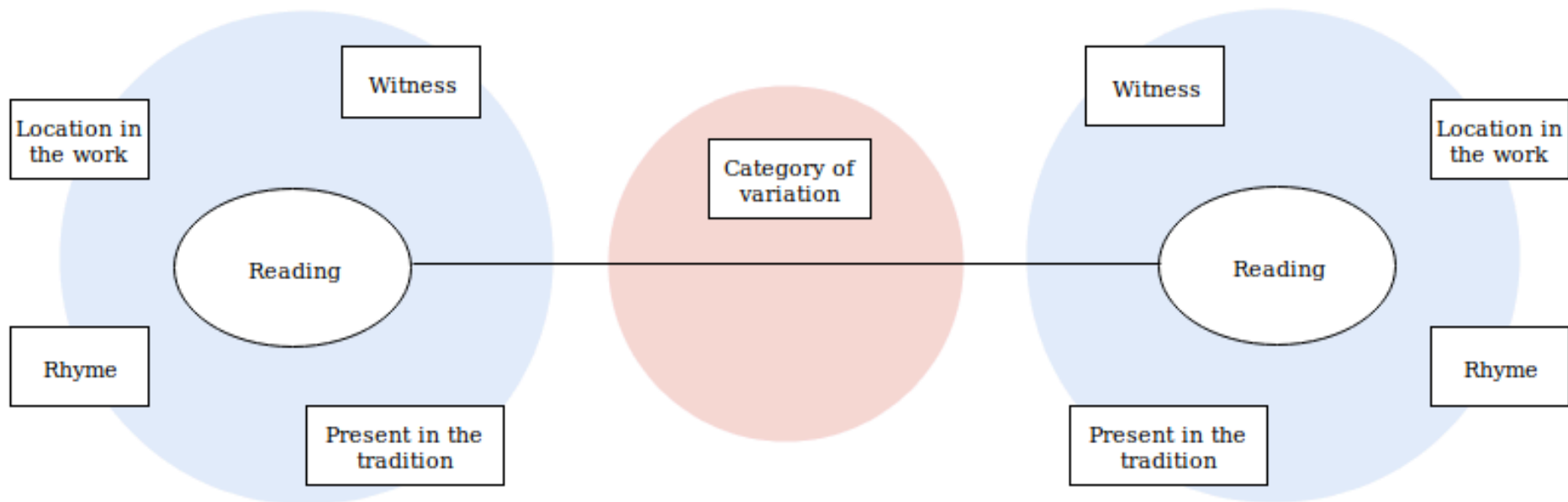
Risultati: 6

In **azzurro** le lezioni assenti nella tradizione precedente a Boccaccio.

Mostra **25** risultati per pagina

Cerca:

			Toledano	Riccardiano	Chigiano	Tipo di variante	In rima	Ed. Petrocchi	
Inf	IV	141	tullio et lyno	tulio lino	tulio lino	inversione	non in rima	Tulïo e Lino e Seneca morale;	
Inf	VIII	24	tal si fece	fecesi tal	fecesi tal	inversione	non in rima	fecesi Flegïas ne l'ira accolta	Nota
Inf	VIII	11	che si et no (ad marg. al. chel si el no)	chel no el si	chel no el si	inversione	non in rima	che si e no nel capo mi tenciona	
Inf	IX	75	ove e quel fummo piu	onde quel fummo e piu	onde quel fummo e piu	inversione	non in rima	per indi ove quel fummo è più acerbo	
Inf	XXXII	135	dissio perche	perche dissio	perche dissio	inversione	non in rima	dimmi 'l perché", diss'io, "per tal convegno,	
Par	IV	47	gabriel et michel	michele et gabriel	michele et gabriel	inversione	non in rima	Gabriel e Michel vi rappresenta,	



THEORY

By "modeling" I mean the heuristic process of constructing and manipulating models, a "model" I take to be either a representation of something for purposes of study, or a design for realizing something new.

[...] modeling has two phases: first, construction; second, manipulation. Examples come readily to mind from ordinary technical practice, e.g., building a relational database, then querying the data thus shaped to explore emergent patterns.

McCarty 2004

Analytical models, common in some scientific practices, necessary for computing -> computational models.

EXERCISE

1

You are creating a database of artworks from a museum exhibition.

In groups, define at least 3 descriptors (characteristic, quality, feature, aspect, information...) to describe and represent each artwork (video sculptures) in the database.

Nam June Paik

- TV Buddha (1974)
- Family of Robot: Baby (1986)
- Internet Dream (1994)

2

You are creating a digital edition of Van Gogh's letters.

In groups, define at least 3 descriptors (characteristic, quality, feature, aspect, information...) to describe and represent each letter in the edition.

Tolworth 15 Nov. 1876

22

Waarde Theo,

Dank voor uw laatste brief dien ik onlangs begelyk met een niet Etten. - Gij zijt dus weer in de zaak doe Gij wat uw hand vindt om te doen met al uw kracht en de regen op werken en bidden zal niet achter blijven. - Wat had ik graag die locht nam het Heleken nam Ipreud el myge maakt in de eerste omeers - Maar vóór ik verder ga schrijf ik een paar verzen over die Gij wel mooi zal vinden.

The journey of life

Two lovers by a mossy grown spring They leaned soft cheeks together Along the dark and sunny hair, And heard the wooing thrushes sing O budding time O loves best prime	Two parents by the evening fire The red light fell about their knees On heads that rose by slow degrees Like buds upon the thistle spike O patient life O tender shift
Two wedded from the postal steps The little maid a happy carolings The air was soft as fanning wings While petals on the path were swept O pure eyed bride O tender pride	The two still sat together there The red light shone about their knees But all the lead by slow degrees Had gone and left that lonely pair O Voyage fast O diminished past
Two faces in a window bent Two hands above the heads were bunched The air was soft as fanning wings These pressed each other while they looked These watched a life which love had rent O solemn hour O hidden power	The red light shone upon the floor And made the space between them wide They drew their chairs up side by side Their pale cheeks joined, and said, <i>more</i> O memories! O past that is!

The three little chairs

They sat alone by the bright window The grey haired woman and the aged man Dreaming of days gone by The hair deep felt in the wrinkled cheek They both had thought that they would not speak And each heart beat a right	But she quivered: Father no, not yet, And look at that, and I forget That the children went away The boys come back, and my Mary too, With her upon an of checkered blue And sit here every day.
For their sad and tearful eyes seemed Three little chairs placed side by side As round the things from walls The fashions enough as they they stood Their seats of fluff, and their frames of wood With their backs so straight and tall	Johnny still whistles as lips hall marks, And wattle his laden bullock cart While Mary her patchwork sews; At evening time three childish prayers Go up to God from those little chairs, So softly that no one knows.
Then the fire took his fiery head And with him long voice he gently said, O mother, those empty chairs, They bring us such sad, sad thoughts long We'll put them for ever out of sight In the small dark room upstairs.	Johnny comes back from the billowy deep, Will not wake from his little bed, sleep, To say good night to me; Mary's wife and mother no more, But a bird child whose play time is And comes to rest on my knee.

het hun licht geleuen dat Gij ook nog eens te
Parys komt. s'avond half 11 was ik weer hier
kerig, ik ging gedachtelijk met de ondergrond
railway terug. - Gelukkig had ik wat geld binnen
gekreuen van Mr Jones. Ben bezig aan Po. 42:1
myne ziel dorst naar God, naar den levenden God
Te Petersham zei ik tot de gemeente dat zij slecht
engelsch konden hooren, maar dat als ik sprak
ik dacht aan den man in de gelijkenis die
zei. heb geduld met my en ik zal u alles
betalen" God helpe mij. - of lieve schets

By Mr. Olbach zag ik het scheldery van
Boughton: the pilgrim progress. - Als Gij ooit
eens kunt krijgen Bunyan's Pilgrim pro-
gress het is zeer de moeite waard om dat
te lezen. Ik voor mij houd er veel van
Het is in den nacht ik zit nog wat te werken
voor de Gladwell te Lewisham, een en ander over
te schrijven en z, men moet het zeer omden als
het heet is en het hart des menschen als het is
brandende en on. - Morgen weer naar Londen
voor Mr Jones. Onder dat van The journey of life
in the blue little chair een mens nog moeten
schryven: Om in de beoeliening van de volheid der
tyden wederom alles tot een te brengen in
Christus, beide dat in den Hemel is en dat op aarde
is. - Zou zij het. - een hand rust in gedachten
grat o. Mr. en Mrs. Terstug voor my en allen by Ross
en Hammebach en o Stockholm. - *more*

gelooft my
in uw liefh. brief
Vincent



THEORY

Il ruolo del modello nella scienza e nel sapere (Roma, 27-28 ottobre 1998), Sergio Carrà (ed.), Accademia nazionale dei Lincei, Roma 1999.

Models [...] represent constructs with a more limited meaning than laws and scientific theories and have an instrumental connotation. In other words, it is generally believed that **the model does not have general applicability, but is limited to a specific context, and can be judged "good" when it proves useful for specific purposes (descriptive, explanatory, or predictive), not necessarily when it is considered "true."**

Maria Carla Galavotti, “Leggi, modelli causali e manipolabilità”

The role of models in science is adequately explained by the analytical model. This is because **formulating a model of certain phenomena is equivalent to formulating a hypothesis about those phenomena**, and the formulation of the model is based on the same logical procedures, such as induction and analogy, that the analytical model uses as the basis for the formulation of hypotheses.

Carlo Cellucci, “I modelli, l'analogia e la metafora”

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Introduction to Data Modelling in the Humanities

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